A DIRECTED COLIMIT OF AFFINE FORMAL ALGEBRAIC SPACES GLUED BY CLOSED IMMERSIONS IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one reindexing of colimits; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Affine formal algebraic spaces as filtered colimits of affine schemes with thickenings as transition maps. Same chapter, Definition definition-affine-formal-algebraic-space, used at proof lines 7876-7880. Supplies the presentations of the individual members.
- Morphisms from a quasi-compact object into a filtered colimit factor through a stage. Same chapter, Lemma lemma-factor-through-thickening, invoked at proof lines 7881-7883. Produces the compatible choices of stages that make the reindexing work.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Suppose given a directed set Λ and a system of affine formal algebraic spaces $(X_\lambda, f_{\lambda\mu})$ over Λ where each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a closed immersion inducing an isomorphism $X_{\lambda,red} \to X_{\mu,red}$. Then $X = \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$ is an affine formal algebraic space over S.*

Proof. We may write $X_\lambda = \operatorname{colim}_{\omega\in\Omega_\lambda} X_{\lambda,\omega}$ as the colimit of affine schemes over a directed set Ω_λ such that the transition morphisms $X_{\lambda,\omega} \to X_{\lambda,\omega'}$ are thickenings. For each $\lambda, \mu \in \Lambda$ and $\omega \in \Omega_\lambda$, with $\mu \geq \lambda$ there exists an $\omega' \in \Omega_\mu$ such that the morphism $X_{\lambda,\omega} \to X_\mu$ factors through $X_{\mu,\omega'}$, see Lemma 0AIA. Then the morphism $X_{\lambda,\omega} \to X_{\mu,\omega'}$ is a closed immersion inducing an isomorphism on reductions and hence a thickening. Set $\Omega = \coprod_{\lambda\in\Lambda} \Omega_\lambda$ and say $(\lambda, \omega) \leq (\mu, \omega')$ if and only if $\lambda \leq \mu$ and $X_{\lambda,\omega} \to X_\mu$ factors through $X_{\mu,\omega'}$. It follows from the above that Ω is a

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directed set and that $X = \operatorname{colim}_{\lambda \in \Lambda} X_\lambda = \operatorname{colim}_{(\lambda,\omega) \in \Omega} X_{\lambda,\omega}$. This finishes the proof. $\square$